Econ 604 Notes

Parker Howell

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Noncooperative Bargaining

Model "Divide a dollar"

- Two players simultaneously submit demands $d_i \in [0, 1]$
- Continuum of Nash equilibria–this will lead us to look at SPE instead.

Model "Ultimatum Game"

- "sequential" but only one round (P1 makes TIOLI offer to P2 who accepts or rejects)
- **Prop:** has a **unique** SPE in which P1 offers (1,0) and P2 accepts any offer.
- multiple NE in the subgame following the offer, but P1 only has a BR if P2 accepts (1,0).

Model "Finite Alternating-Offer Bargaining"

Regularities:

1. Immediate agreement, efficient outcome
2. Proposer's offer makes responder indifferent
3. Unique SPE
4. Division of surplus depends:
    1. $x_1^*(T, \delta) = \sum_{t=1}^{T} (-\delta)^{t-1} = \frac{1-(-\delta)^T}{1+\delta}$
    2. $\delta \to 1 \Rightarrow$ last mover takes all
    3. $T \to \infty \Rightarrow X^* \to \frac{1}{1+\delta}$
    4. $T \to \infty$ then $\delta \to 1 \Rightarrow$ equal sharing

    Every game in this class has a unique subgame perfect equilibrium, and in it there is immediate agreement in the first round.

    The proposer in round T offers to take everything, and the responder accepts every offer

Thm "infinite alternating-offer bargaining game outcome"

The infinite horizon alternating-offer bargaining game with discount factor $\delta < 1$ has a unique subgame perfect equilibrium, in which Player 1 offers the payoff vector $(\frac{1}{1+\delta}, \frac{\delta}{1+\delta})$ in the first round and Player 2 accepts.

Repeated Games

Finitely Repeated Games

Model "Repeated Games"

- Stage Game $\langle N, A, \pi \rangle$ (Normal form game)
- Players $N = \{1, ..., N\}$

- Action profiles $A = A_1 \times \cdots A_n$
- Stage payoffs $\pi : A \to \mathbb{R}^n$ (uniformly bounded) (i.e., $\pi_i(a)$ is i's payoff from action profile a)
- Periods $t = 1, ..., T$, where $T < \infty$
- Discount factor $\delta > 0 (\delta < 1$ if $T = \infty)$

"Histories and Strategies"

A strategy (in a repeated game) maps from the history of past play into an action for the current period.

- History $h^t \equiv (a^1, ..., a^{t-1})$, $h^1 \equiv \varnothing$
- Possible t-histories $\mathcal{H} \equiv A^{t-1}$
- Possible histories $\mathcal{H} \equiv \cup_{t=1}^{T} \mathcal{H}^t$
- Correlated behavior strategy profile $\sigma : \mathcal{H} \to \Delta A$
- A (pure) strategy is a function $s_i : \mathcal{H} \to A_i$
- a (behavioral) mixed strategy is defined in terms of strategy profiles to allow for correlation. So $\sigma : \mathcal{H} \to \Delta A$.

  a behavioral mixed-strategy must specify not only what players play "sequentially", but also in all other possible histories of the game.

Def "Utility"

$$\text{Total Utility} = \tilde{u}_i(\sigma) \equiv \sum_{t=1}^{T} \delta^{t-1} \sum_{h^t \in \mathcal{H}^t} \pi_i(\sigma(h^t)) Pr(h^t|\sigma)$$

$$\text{Average Utility} = u_i(\sigma) \equiv \frac{1-\delta}{1-\delta^T} \sum_{t=1}^{T} \delta^{t-1} \sum_{h^t \in \mathcal{H}^t} \pi_i(\sigma(h^t)) Pr(h^t|\sigma)$$

For infinite horizon we multiply by $1 - \delta$, which is just the limit as $T \to \infty$ of $\frac{1-\delta}{1-\delta^T}$.

Infinitely Repeated Games

Def "Non-contingent strategy profile"

σ is a non-contingent strategy profile if

$$\sigma(h^t) = \sigma(\hat{h}^t), \forall h^t, \hat{h}^t \in \mathscr{H}^t, \forall t = 1, ..., T.$$

σ is non-contingent if it specifies that in each period players should play some period-specific mixed action profile regardless of the history.

Prop 13.1 "Non-contingent equilibria"

Suppose $T < \infty$. If σ is a non-contingent strategy profile such that $\sigma(h^t)$ is a stage game NE for all $t = 1, ..., T$ then σ is a SPE.

Any sequence of stage Nash profiles can be supported as the outcome of a SPE. (Watson)

Prop 13.2 "Unique Stage Game Equilibrium"

Suppose $T < \infty$. If α^* is the unique NE in the stage game, then the unique SPE in the repeated game is a non-contingent strategy profile σ^* such that, for all $h \in \mathscr{H}, \sigma^*(h) = \alpha^*$.

Def "profitable one-shot deviation"

A *profitable one-shot deviation* from σ is a strategy $\hat{\sigma}_i \neq \sigma_i$ for some player i such that:

1. there exists a unique history $\hat{h}^t \in \mathscr{H}$ such that $\hat{\sigma}_i(h) = \sigma_i(h), \forall h \neq \hat{h}$ and
2. $\tilde{u}_i(\hat{\sigma}_i|_{\hat{h}}, \sigma_{-i}|_{\hat{h}}) > \tilde{u}_i(\sigma_i|_{\hat{h}})$.

Thm "one-shot deviation principle"

In a finitely or infinitely repeated game, a strategy profile is a SPE if and only if it has no profitable one-shot deviations.

Model "Infinite Cournot Duopoly"

- Two firms simultaneously choose *quantities* in each period
- Demand curve (e.g., $P = 12 - Q$)
- Zero costs
- $\delta < 1$

Proposal:

- on path: produce 1/2 monopoly quantity each
- off path: produce Cournot quantity (i.e., stage game NE)

Q. For what value of δ is this proposal a SPE?

Model "Infinite Bertrand Duopoly"

- Two firms simultaneously choose *prices* in each period
- Unit mass of consumers demand 1 unit at any price ≤ 1
- lowest price gets the market; ties split
- Zero costs
- $\delta < 1$

Proposal:

- on-path: set price = 1 (note: this means market is split)
- off-path: set price = 0

Q. For what value of δ is this proposal a SPE?

The Folk Theorem

Thm "Nash-Threat Folk Theorem"

If v is feasible and Nash-individually rational, then there exists $\underline{\delta} < 1$ such that, for all $\delta \in (\underline{\delta}, 1)$, v is the average utility profile of an SPE given δ.

Def "feasible"

A payoff vector $v \in \mathbb{R}^n$ is *feasible* if v is in the convex hull of $\{\pi(a) : a \in A\}$

Def "Nash-individually rational"

A payoff vector $v \in \mathbb{R}^n$ is *Nash-individually rational* if

$$v_i > \underline{v}_i^{Nash} \equiv \min_{\alpha \in \Delta A \text{ is Nash}} \pi_i(\alpha), \forall i.$$

Def "Minimax-individually rational"

A payoff vector $v \in \mathbb{R}^n$ is *individually rational* if

$$v_i > \underline{v}_i^{min} \equiv \min_{\alpha_{-i} \in \Delta A_{-i}} \max_{a_i \in A_i} \pi_i(a_i, \alpha_{-i}), \forall i.$$

Remark $\underline{v}_i^{min} \leq \underline{v}_i^{Nash}, \forall i$

Thm "Minimax Folk Theorem"

If v is in the interior of the set of feasible and individually rational payoffs, then there exists $\underline{\delta} < 1$ such that, for all $\delta \in (\underline{\delta}, 1)$, v is the average utility profile of an SPE given δ.

Bayesian Nash Equilibrium

Bayesian Games

Def "Bayesian Game"

A Bayesian game is a tuple $\langle N, A, u, \Theta, \mu \rangle$ where

- each player i privately observes some signal $\theta_i \in \Theta$
- Nature draws a type vector $\theta = (\theta_1, ..., \theta_n) \in \Theta$ from a common prior measure μ on Θ
- $u_i : A \times \Theta \to \mathbb{R}$

A strategy for player i is a measurable function mapping from Θ_i to ΔA_i. S_i is the space of such functions.

Def "Bayesian Normal Form"

$\langle N, S, \mathbb{E}[u] \rangle$ where $\mathbb{E}[u_i(s)] \equiv \mathbb{E}[u_i(s(\theta), \theta)]$ is player i's expected payoff given strategy profile s and the common prior μ.

Def "Bayesian Nash Equilibrium"

In a Bayesian game, $\langle N, A, u, \Theta, \mu \rangle$, a *Bayesian Nash Equilibrium* is a strategy profile $s \in S = \times_{i=1}^{n} S_i$ such that for each i and each $s_i' \in S_i$,

$$\mathbb{E}[u_i(s(\theta), \theta)] \geq \mathbb{E}[u_i(s_i'(\theta_i), s_{-i}(\theta_{-i}), \theta)].$$

Observation

A BNE is a NE in the game $\langle N, S, \mathbb{E}[u] \rangle$.

Thm "Bayesian Nash Equilibrium if"

In a Bayesian game, $\langle N, A, u, \Theta, \mu \rangle$, a strategy profile $s \in S = \times_{i=1}^{n} S_i$ is a *Bayesian Nash Equilibrium* if

$$\mathbb{E}[u_i(s(\theta), \theta | \theta_i)] \geq \mathbb{E}[u_i(a_i, s_{-i}(\theta_{-i}) | \theta_i)], \forall a_i \in A_i, \theta_i \in \Theta_i, i \in N.$$

Remark This is also "only if" if Θ is finite and μ has full support.

Applications

Model "Cournot Duopoly with higher order beliefs"

- 2 firms simultaneously choose quantities facing a demand curve
- Firm 1 has zero costs
- Firm 2 is
  - type $\theta_2 = L$ (i.e., has low costs) with probability $1 - \alpha$
  - type $\theta_2 = H$ (i.e., has high costs) with probability α
- Firm 1's belief about Firm 2's costs:
  - Type $\theta_1 = l$ (i.e., believes $\alpha_l = 1/4$) with probability $1 - \beta$
  - Type $\theta_1 = h$ (i.e., believes $\alpha_h = 3/4$) with probability β

Model "Lemons"

- Seller privately knows his good has quality $\theta \in [0,1]$
- Buyer believes $\theta \sim U[0,1]$
- Seller chooses price $p \in [0,1]$
- Buyer chooses set $E \subseteq [0,1]$ of prices she will accept

Analysis:

1. $E^* = \emptyset \Rightarrow$ continuum of no-trade equilibria
2. $E^* \neq \emptyset \Rightarrow E^*$ contains its maximum, e^*
3. Seller BR to e^*: $BR_s(e^*|\theta) = \begin{cases} e^* \text{ if } \theta < \frac{e^*}{q} \\ (e^*, 1] \text{ if } \theta > \frac{e^*}{q} \end{cases}$.
4. $E^* \neq \emptyset \Rightarrow e^* = 0$
5. $E^* = \{0\} \Rightarrow$ continuum of almost-no-trade equilibria

Term: "Adverse selection" Occurs when asymmetric information leads to undesirable outcomes in transactions.

Auctions

Primitives

- n bidders (one prize)
- Bidder i's type: $\theta_i \in \Theta_i$
- Each bidder simultaneously submits $b_i \in [0, \infty]$
- Payoff: $u_i(b, \theta) = x_i(b)v_i(\theta) - p_i(b)$
  - $x_i(b)$ is the probability i gets the prize
  - $v_i(b)$ is i valuation of the prize
  - $p_i(b)$ is the price i pays

Model "Second-price *Private* Values"

- Private values: $\Theta_i = [0, \bar{\theta}]$ and $v_i(\theta) = \theta_i$
- **Thm:** It is a weakly dominant strategy for each player to bid $b_i(\theta_i) = \theta_i$, regardless of the distribution over Θ.

Model "First-price *Private* Values"

- (Confirmed) Guess: symmetric strictly increasing bid strategy, $s : \Theta_i \to [0, \infty)$.

Model "Second-price *Common* Values"

- **Claim:** Everyone bidding their true expected values is not an equilibrium (suppose not: bid lower, now you (1) lose if you were close, or (2) profitable deviation??)
- **Claim:** Symmetric equilibrium in which each player bids his expected value conditional on θ_i and $\max_{j \neq i} \theta_j = \theta_i$. (Winner's curse!)